Complete PostgreSQL Schema for SabiLens

Database Setup

sql

-- Create database

CREATE DATABASE sabilens;

\c sabilens;

-- Enable extensions

CREATE EXTENSION IF NOT EXISTS "uuid-ossp";

CREATE EXTENSION IF NOT EXISTS "postgis"; -- For geospatial queries

CREATE EXTENSION IF NOT EXISTS "pgcrypto"; -- For encryption

1. Users & Authentication Schema

```sql

-- Users table (all roles: consumer, company_admin, nafdac_officer, nafdac_admin)

CREATE TABLE users (

  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),

  email VARCHAR(255) UNIQUE,

  phone VARCHAR(20) UNIQUE NOT NULL,

  password_hash VARCHAR(255) NOT NULL,

  first_name VARCHAR(100),

  last_name VARCHAR(100),

  role VARCHAR(50) NOT NULL CHECK (role IN ('consumer', 'company_admin', 'nafdac_officer', 'nafdac_admin')),

  status VARCHAR(50) DEFAULT 'active' CHECK (status IN ('active', 'suspended', 'pending', 'deleted')),

  email_verified BOOLEAN DEFAULT FALSE,

  phone_verified BOOLEAN DEFAULT FALSE,

  profile_image_url TEXT,
```

```
    preferred_language VARCHAR(10) DEFAULT 'en',
    last_login TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    deleted_at TIMESTAMP
);

-- Create index for faster lookups
CREATE INDEX idx_users_email ON users(email) WHERE deleted_at IS NULL;
CREATE INDEX idx_users_phone ON users(phone) WHERE deleted_at IS NULL;
CREATE INDEX idx_users_role ON users(role);

-- Sessions table
CREATE TABLE sessions (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    user_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE,
    refresh_token TEXT NOT NULL,
    device_info JSONB,
    ip_address INET,
    expires_at TIMESTAMP NOT NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_sessions_user_id ON sessions(user_id);
CREATE INDEX idx_sessions_refresh_token ON sessions(refresh_token);

-- OTP verifications
CREATE TABLE otp_verifications (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    user_id UUID REFERENCES users(id) ON DELETE CASCADE,
    phone VARCHAR(20),
    email VARCHAR(255),
    otp_code VARCHAR(6) NOT NULL,
```

```
    purpose VARCHAR(50) CHECK (purpose IN ('registration', 'login', 'password_reset',
'phone_verification')),
    expires_at TIMESTAMP NOT NULL,
    verified_at TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT at_least_one_contact CHECK (phone IS NOT NULL OR email IS NOT NULL)
);

CREATE INDEX idx_otp_phone ON otp_verifications(phone);
CREATE INDEX idx_otp_email ON otp_verifications(email);
```

2. Companies Schema

```sql
-- Companies table
CREATE TABLE companies (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    name VARCHAR(255) NOT NULL,
    registration_number VARCHAR(100) UNIQUE NOT NULL,
    email VARCHAR(255) UNIQUE NOT NULL,
    phone VARCHAR(20) NOT NULL,
    address TEXT,
    city VARCHAR(100),
    state VARCHAR(100),
    country VARCHAR(100) DEFAULT 'Nigeria',
    logo_url TEXT,
    website VARCHAR(255),
    status VARCHAR(50) DEFAULT 'pending' CHECK (status IN ('pending', 'active', 'suspended',
'rejected')),
    subscription_tier VARCHAR(50) DEFAULT 'free' CHECK (subscription_tier IN ('free', 'basic',
'premium', 'enterprise')),
    subscription_expires_at TIMESTAMP,
```

```
    verified_at TIMESTAMP,
    verified_by UUID REFERENCES users(id),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_companies_status ON companies(status);
CREATE INDEX idx_companies_registration ON companies(registration_number);

-- Company team members (users with company_admin role)
ALTER TABLE users ADD COLUMN company_id UUID REFERENCES companies(id) ON DELETE SET
NULL;

-- Company settings
CREATE TABLE company_settings (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    company_id UUID NOT NULL REFERENCES companies(id) ON DELETE CASCADE,
    notification_email BOOLEAN DEFAULT TRUE,
    notification_alerts BOOLEAN DEFAULT TRUE,
    notification_reports BOOLEAN DEFAULT TRUE,
    auto_export_enabled BOOLEAN DEFAULT FALSE,
    api_access_enabled BOOLEAN DEFAULT FALSE,
    webhook_url TEXT,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    UNIQUE(company_id)
);

-- Company API keys
CREATE TABLE company_api_keys (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    company_id UUID NOT NULL REFERENCES companies(id) ON DELETE CASCADE,
    api_key VARCHAR(255) UNIQUE NOT NULL,
```

```
    api_secret VARCHAR(255) NOT NULL,
    name VARCHAR(100) NOT NULL,
    permissions JSONB DEFAULT '[]',
    last_used_at TIMESTAMP,
    expires_at TIMESTAMP,
    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_api_keys_company ON company_api_keys(company_id);
```

3. Products Registry

```sql
-- Product categories
CREATE TABLE product_categories (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    name VARCHAR(100) NOT NULL,
    description TEXT,
    parent_id UUID REFERENCES product_categories(id),
    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- Products
CREATE TABLE products (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    company_id UUID NOT NULL REFERENCES companies(id) ON DELETE CASCADE,
    name VARCHAR(255) NOT NULL,
    description TEXT,
    category_id UUID REFERENCES product_categories(id),
```

```
    nafdac_number VARCHAR(100) UNIQUE NOT NULL,
    batch_number VARCHAR(100),
    manufacturer_name VARCHAR(255),
    manufacturing_date DATE,
    expiry_date DATE,
    ingredients TEXT,
    warnings TEXT,
    image_urls TEXT[],
    master_artwork_url TEXT,
    status VARCHAR(50) DEFAULT 'active' CHECK (status IN ('active', 'pending', 'expired',
'discontinued')),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    created_by UUID REFERENCES users(id)
);

CREATE INDEX idx_products_company ON products(company_id);
CREATE INDEX idx_products_nafdac ON products(nafdac_number);
CREATE INDEX idx_products_status ON products(status);

-- Product barcodes
CREATE TABLE product_barcodes (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    product_id UUID NOT NULL REFERENCES products(id) ON DELETE CASCADE,
    barcode_type VARCHAR(50) CHECK (barcode_type IN ('UPC', 'EAN', 'QR', 'CODE128', 'OTHER')),
    barcode_value VARCHAR(255) NOT NULL,
    is_primary BOOLEAN DEFAULT FALSE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    UNIQUE(barcode_type, barcode_value)
);
```

4. Scan History & Reports

```sql
-- Scans table
CREATE TABLE scans (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    user_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE,
    product_id UUID REFERENCES products(id),
    product_name VARCHAR(255),
    category VARCHAR(100),
    manufacturer VARCHAR(255),
    nafdac_number VARCHAR(100),
    batch_number VARCHAR(100),
    expiry_date DATE,
    status VARCHAR(50) CHECK (status IN ('authentic', 'caution', 'fake', 'pending')),
    similarity_score DECIMAL(5,2),
    confidence_score DECIMAL(5,2),
    scan_image_urls TEXT[],
    scan_location GEOGRAPHY(POINT),
    location_name VARCHAR(255),
    scan_metadata JSONB,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_scans_user ON scans(user_id);
CREATE INDEX idx_scans_product ON scans(product_id);
CREATE INDEX idx_scans_status ON scans(status);
CREATE INDEX idx_scans_created ON scans(created_at);
CREATE INDEX idx_scans_location ON scans USING GIST(scan_location);

-- Scan analysis results (detailed AI output)
CREATE TABLE scan_analyses (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
```

```
    scan_id UUID NOT NULL REFERENCES scans(id) ON DELETE CASCADE,
    ai_model_version VARCHAR(50),
    features_detected JSONB,
    anomalies JSONB,
    ocr_text TEXT,
    visual_fingerprint JSONB,
    analysis_time_ms INTEGER,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- Counterfeit reports
CREATE TABLE reports (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    scan_id UUID NOT NULL REFERENCES scans(id) ON DELETE CASCADE,
    user_id UUID NOT NULL REFERENCES users(id),
    product_id UUID REFERENCES products(id),
    company_id UUID REFERENCES companies(id),
    report_status VARCHAR(50) DEFAULT 'pending' CHECK (report_status IN ('pending', 'verified',
'investigating', 'resolved', 'rejected')),
    severity VARCHAR(50) CHECK (severity IN ('critical', 'high', 'medium', 'low')),
    purchase_channel VARCHAR(50) CHECK (purchase_channel IN ('market_stall', 'roadside_vendor',
'supermarket', 'supplier', 'online', 'other')),
    purchase_channel_details TEXT,
    price_paid DECIMAL(10,2),
    receipt_image_urls TEXT[],
    additional_comments TEXT,
    gps_location GEOGRAPHY(POINT),
    location_name VARCHAR(255),
    reported_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    resolved_at TIMESTAMP,
    resolved_by UUID REFERENCES users(id),
    resolution_notes TEXT
);
```

```
CREATE INDEX idx_reports_scan ON reports(scan_id);
CREATE INDEX idx_reports_company ON reports(company_id);
CREATE INDEX idx_reports_status ON reports(report_status);
CREATE INDEX idx_reports_severity ON reports(severity);
CREATE INDEX idx_reports_location ON reports USING GIST(gps_location);

-- Report evidence
CREATE TABLE report_evidence (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    report_id UUID NOT NULL REFERENCES reports(id) ON DELETE CASCADE,
    evidence_type VARCHAR(50) CHECK (evidence_type IN ('image', 'video', 'receipt', 'document')),
    file_url TEXT NOT NULL,
    file_size INTEGER,
    mime_type VARCHAR(100),
    uploaded_by UUID REFERENCES users(id),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

5. Alerts & Notifications

```sql
-- Alerts (for companies and NAFDAC)
CREATE TABLE alerts (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    report_id UUID REFERENCES reports(id) ON DELETE SET NULL,
    company_id UUID REFERENCES companies(id) ON DELETE CASCADE,
    alert_type VARCHAR(50) CHECK (alert_type IN ('counterfeit', 'suspicious', 'bulk', 'recurring')),
    severity VARCHAR(50) CHECK (severity IN ('critical', 'high', 'medium', 'low')),
    title VARCHAR(255) NOT NULL,
    message TEXT NOT NULL,
```

```
    metadata JSONB,
    is_read BOOLEAN DEFAULT FALSE,
    is_acknowledged BOOLEAN DEFAULT FALSE,
    acknowledged_by UUID REFERENCES users(id),
    acknowledged_at TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_alerts_company ON alerts(company_id);
CREATE INDEX idx_alerts_type ON alerts(alert_type);
CREATE INDEX idx_alerts_severity ON alerts(severity);
CREATE INDEX idx_alerts_created ON alerts(created_at);

-- Notifications (for users)
CREATE TABLE notifications (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    user_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE,
    notification_type VARCHAR(50) CHECK (notification_type IN ('scan_result', 'report_update',
'alert', 'system', 'achievement')),
    title VARCHAR(255) NOT NULL,
    message TEXT NOT NULL,
    data JSONB,
    is_read BOOLEAN DEFAULT FALSE,
    read_at TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_notifications_user ON notifications(user_id);
CREATE INDEX idx_notifications_read ON notifications(is_read);

-- User notification preferences
CREATE TABLE notification_preferences (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
```

```
    user_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE,
    notification_type VARCHAR(50) NOT NULL,
    channel VARCHAR(50) CHECK (channel IN ('push', 'email', 'sms', 'in_app')),
    enabled BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    UNIQUE(user_id, notification_type, channel)
);
```

6. Geospatial & Heat Map Data

```sql
-- Counterfeit hotspots
CREATE TABLE hotspots (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    location GEOGRAPHY(POINT) NOT NULL,
    region_name VARCHAR(255),
    city VARCHAR(100),
    state VARCHAR(100),
    country VARCHAR(100) DEFAULT 'Nigeria',
    report_count INTEGER DEFAULT 0,
    severity_score DECIMAL(5,2),
    risk_level VARCHAR(50) CHECK (risk_level IN ('critical', 'high', 'medium', 'low')),
    last_updated TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_hotspots_location ON hotspots USING GIST(location);
CREATE INDEX idx_hotspots_risk ON hotspots(risk_level);

-- Vendor locations (where counterfeits were purchased)
```

```
CREATE TABLE vendor_locations (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    report_id UUID REFERENCES reports(id) ON DELETE SET NULL,
    vendor_name VARCHAR(255),
    location GEOGRAPHY(POINT) NOT NULL,
    address TEXT,
    city VARCHAR(100),
    state VARCHAR(100),
    report_count INTEGER DEFAULT 1,
    risk_level VARCHAR(50) DEFAULT 'unknown' CHECK (risk_level IN ('critical', 'high', 'medium', 'low', 'unknown')),
    is_blacklisted BOOLEAN DEFAULT FALSE,
    blacklisted_at TIMESTAMP,
    blacklisted_by UUID REFERENCES users(id),
    last_reported_at TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_vendors_location ON vendor_locations USING GIST(location);
CREATE INDEX idx_vendors_risk ON vendor_locations(risk_level);
CREATE INDEX idx_vendors_blacklisted ON vendor_locations(is_blacklisted);
```

7. Analytics & Statistics

```sql
-- Daily/Monthly aggregates
CREATE TABLE analytics_daily (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    date DATE NOT NULL,
    total_scans INTEGER DEFAULT 0,
```

```
    authentic_scans INTEGER DEFAULT 0,
    suspicious_scans INTEGER DEFAULT 0,
    counterfeit_scans INTEGER DEFAULT 0,
    total_reports INTEGER DEFAULT 0,
    unique_users INTEGER DEFAULT 0,
    unique_companies INTEGER DEFAULT 0,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    UNIQUE(date)
);

-- Category analytics
CREATE TABLE analytics_categories (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    category_id UUID REFERENCES product_categories(id),
    category_name VARCHAR(100),
    period VARCHAR(20) CHECK (period IN ('daily', 'weekly', 'monthly', 'quarterly')),
    period_start DATE NOT NULL,
    period_end DATE NOT NULL,
    scan_count INTEGER DEFAULT 0,
    counterfeit_count INTEGER DEFAULT 0,
    suspicious_count INTEGER DEFAULT 0,
    authentic_count INTEGER DEFAULT 0,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- Regional analytics
CREATE TABLE analytics_regions (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    state VARCHAR(100) NOT NULL,
    city VARCHAR(100),
    period VARCHAR(20) CHECK (period IN ('daily', 'weekly', 'monthly', 'quarterly')),
    period_start DATE NOT NULL,
    period_end DATE NOT NULL,
```

```
  scan_count INTEGER DEFAULT 0,
  counterfeit_count INTEGER DEFAULT 0,
  risk_score DECIMAL(5,2),
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

8. Enums & Lookup Tables

```sql
-- Nigerian states
CREATE TABLE nigerian_states (
  id SERIAL PRIMARY KEY,
  name VARCHAR(100) UNIQUE NOT NULL,
  capital VARCHAR(100),
  region VARCHAR(50)
);

INSERT INTO nigerian_states (name, region) VALUES
('Abia', 'South East'),
('Adamawa', 'North East'),
('Akwa Ibom', 'South South'),
('Anambra', 'South East'),
('Bauchi', 'North East'),
('Bayelsa', 'South South'),
('Benue', 'North Central'),
('Borno', 'North East'),
('Cross River', 'South South'),
('Delta', 'South South'),
('Ebonyi', 'South East'),
('Edo', 'South South'),
('Ekiti', 'South West'),
```

```
('Enugu', 'South East'),
('FCT', 'North Central'),
('Gombe', 'North East'),
('Imo', 'South East'),
('Jigawa', 'North West'),
('Kaduna', 'North West'),
('Kano', 'North West'),
('Katsina', 'North West'),
('Kebbi', 'North West'),
('Kogi', 'North Central'),
('Kwara', 'North Central'),
('Lagos', 'South West'),
('Nasarawa', 'North Central'),
('Niger', 'North Central'),
('Ogun', 'South West'),
('Ondo', 'South West'),
('Osun', 'South West'),
('Oyo', 'South West'),
('Plateau', 'North Central'),
('Rivers', 'South South'),
('Sokoto', 'North West'),
('Taraba', 'North East'),
('Yobe', 'North East'),
('Zamfara', 'North West');

-- Languages
CREATE TABLE languages (
   id SERIAL PRIMARY KEY,
   code VARCHAR(10) UNIQUE NOT NULL,
   name VARCHAR(50) NOT NULL,
   is_active BOOLEAN DEFAULT TRUE
);
```

```
INSERT INTO languages (code, name) VALUES
('en', 'English'),
('yo', 'Yorùbá'),
('ha', 'Hausa'),
('ig', 'Igbo'),
('pcm', 'Pidgin');
```

9. Audit & Logs

```sql
-- Audit logs (for compliance)
CREATE TABLE audit_logs (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    user_id UUID REFERENCES users(id),
    action VARCHAR(100) NOT NULL,
    entity_type VARCHAR(50),
    entity_id UUID,
    old_data JSONB,
    new_data JSONB,
    ip_address INET,
    user_agent TEXT,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_audit_user ON audit_logs(user_id);
CREATE INDEX idx_audit_entity ON audit_logs(entity_type, entity_id);
CREATE INDEX idx_audit_created ON audit_logs(created_at);

-- System logs
CREATE TABLE system_logs (
```

```
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    log_level VARCHAR(20) CHECK (log_level IN ('debug', 'info', 'warning', 'error', 'critical')),
    component VARCHAR(100),
    message TEXT,
    metadata JSONB,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

10. Relationships Summary

```sql
-- Users belong to Companies (if company_admin)
-- Companies have many Products
-- Users have many Scans
-- Scans generate Reports
-- Reports belong to Companies and Products
-- Reports generate Alerts for Companies and NAFDAC
-- Reports create Vendor Locations
-- Locations create Hotspots
-- All actions are Audited
```